1. PACT Analysis

Proposed System: A touchless media controller that uses a camera to recognize hand gestures, allowing users to control media playback without physically touching their devices.

PACT Element	Analysis
P - People	People who watch videos or listen to music while cooking, washing dishes, eating, or performing other activities that make their hands dirty, wet, or occupied. Users may have different levels of technical experience.
A - Activities	Playing or pausing media, adjusting volume, and skipping songs or videos without touching the device. These are short, frequent interactions that should require minimal effort.
C - Context	The system is mainly used in kitchens, dining areas, or workspaces. Users may be standing 3-5 feet from their device, with background noise, varying lighting, and limited attention. Their hands might be wet, greasy, or dirty.
T - Technology	A laptop or computer with a camera, a computer vision model for recognizing hand gestures, and software that translates gestures into media controls. The system provides visual feedback when a gesture is recognized.

2. Realistic User Scenario

Scenario Title: Controlling YouTube While Washing Dishes
User: Ahmed, a 22-year-old university student.
Location: Home kitchen.
Goal: Control YouTube playback without touching his laptop with wet hands.

Scenario:
Ahmed enjoys watching YouTube videos while washing dishes. He places his laptop on the kitchen counter, approximately four feet away from the sink, and enables the gesture-control feature before starting.

Step 1 - Watching a video
Ahmed starts washing dishes while watching a YouTube video on his laptop. His hands become wet and soapy.

Step 2 - Pausing the video
Ahmed needs to answer a phone call. Instead of drying his hands, he raises his open palm toward the laptop camera and holds it briefly. The system recognizes the gesture, pauses the video, and displays a pause icon.

Step 3 - Resuming playback
After finishing the call, Ahmed repeats the open-palm gesture. The video resumes playing without him touching the laptop.

Step 4 - Increasing the volume
The sound of running water makes the video difficult to hear. Ahmed moves his hand upward in front of the camera. The system increases the volume and displays the updated volume level.

Outcome: Ahmed successfully controls his video without drying his hands, touching the laptop, or interrupting his dishwashing activity.

3. Survey

Target Users

People who use media apps such as YouTube, Spotify, and Netflix while their hands are dirty, wet, or occupied.

Purpose

To understand users' difficulties controlling media without touching their devices, their preferred gestures, and their comfort with camera-based gesture recognition.

1. How often do you interact with media while your hands are dirty, wet, or otherwise occupied?

- Multiple times a day
- Once a day
- A few times a week
- Rarely or never

Why it's important: Helps us understand how frequently users experience the problem.

2. In which scenarios do you most frequently encounter this issue?

- Cooking or baking
- Cleaning or washing dishes
- Arts, crafts, or DIY projects
- Mechanics, gardening, or heavy manual work
- Eating messy finger foods
- Other: _____

Why it's important: Identifies situations where touch-free controls could be useful.

3. How do you control your media when your hands are dirty?

- Wipe my hands on a towel or apron.
- Use my nose, knuckle, or elbow to tap the screen.
- Use voice commands.
- Let the media play without changing it.
- Other: _____

Why it's important: Shows how users currently deal with the problem and which alternatives they use.

Critique: The question misses alternatives such as using a remote or waiting until their hands are free.

4. Which problems do you experience with your current method?

- Smudges, grease, or water on my screen.
- Voice assistants not hearing me over background noise.
- The effort needed to clean my hands.
- Voice assistants interrupting the media.
- I don't experience any major frustrations.

Why it's important: Identifies the main problems the project should solve.

Critique: "I don't experience any major frustrations" should be selected on its own to avoid conflicting answers.

5. If you were 3–5 feet away from your screen, which hand gesture would feel most natural for pausing a video?

- Holding up an open palm facing the camera.
- Putting a finger to my lips.
- Making a fist.
- Waving one hand side-to-side.
- Other: _____
- None of these.

Why it's important: Helps us choose a pause gesture that users find natural.

Critique: Adding "None of these" prevents users from being forced to choose a gesture they dislike.

6. Which movement would feel most natural for turning up the volume without touching the device?

- Rotating my hand clockwise.
- Moving an open hand or finger upward.
- Swiping my hand to the right.
- Making a thumbs-up sign.
- Other: _____
- None of these.

Why it's important: Helps us identify an intuitive gesture for volume control.

7. How comfortable would you feel using your device's camera to recognize hand gestures for controlling media?

- Very comfortable
- Somewhat comfortable
- Neutral
- Somewhat uncomfortable
- Very uncomfortable

Why it's important: Measures users' comfort with the proposed technology.

Critique: Comfort may depend on privacy concerns, so explaining whether video is stored would help users answer more accurately.